

Program: Diploma in Automation and Robotics	
Course Code: 5339B	Course Title: Machine Vision Lab
Semester: 5	Credits: 3
Course Category: Program Elective	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To introduce the student to OpenCV, especially for image processing.
- To Calibrate the camera and extract the intrinsic and extrinsic parameters of the camera
- To Find the edges in the image

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

Course Prerequisites:

Topic	Course code	Coursename	Semester
Basicconceptsofpython programming.	3333	Python Programming	3

Tools and Equipment:

Processors: Intel Atom® processor or Intel® Core™ i3 processor.

Disk space: 1 GB.

Operating systems: Windows 7 or later, macOS, and Linux.

Software: PyCharm, OpenCV

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To Perform basic Image Handling and Processing operations on the image.	14	Applying
CO2	To perform various techniques of image enhancement, reconstruction, compression and segmentation	8	Applying
CO3	To perform Edge Detection, Line Detection and Corner Detection.	12	Applying
CO4	To perform Camera Calibration	8	Applying
	Lab Exam	3	

CO–POMapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			
CO5	3			3			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To Perform basic Image Handling and Processing operations on the image.		

M1.01	Reading, displaying, and writing an image using OpenCV.	2	Applying
M1.02	Convert the image to another format using OpenCV	2	Applying
M1.03	Perform the image resizing using OpenCV.	2	Applying
M1.04	Convert a colored image into a grayscale image using OpenCV.	2	Applying
M1.05	Scaling, rotation and shifting operation on the image using OpenCV.	2	Applying
M1.06	Play a video using OpenCV.	2	Applying
M1.07	Extract images from the video.	2	Applying
CO2	To perform various techniques of image enhancement, reconstruction, compression and segmentation		
M2.01	Perform scaling, rotation and shifting operations on an image using OpenCV()	2	Applying
M2.02	Perform image reflection on an image using OpenCV().	2	Applying
M2.03	Perform Image shearing on an image using OpenCV().	2	Applying
M2.04	Apply the affine transformation on an image using OpenCV().	2	Applying
	Lab ExamI	1.5	
CO3	To perform Edge Detection, Line Detection and Corner Detection.		
M3.01	Compute the edge detection using Sobel, Prewitt and canny operator.	3	Applying
M3.02	Implement the Harris Corner detector algorithm to determine the corner in the image.	3	Applying
M3.03	Implement the Harris Corner Detector algorithm without the inbuilt OpenCV() function.	3	Applying
M3.04	Detect the line using Hough Transform	3	Applying

CO4	To perform Read and write data from/to files in Python		
M4.01	Perform the camera calibration and compute the intrinsic and extrinsic parameters of the camera.	8	Applying
	LabExam II	1.5	

Online Resources:

Sl.No	WebsiteLink
1	OpenCV Python Tutorial –GeeksforGeeks
2	OpenCV-python-tests/cameraCalibration.py at master · jagracar/OpenCV-python-tests · GitHub
3	OpenCV: Image Gradients
4	https://github.com/